Advanced Numerical Methods for Partial Differential Equations

A Preprint

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Abstract

This is the abstract for the preprint. It summarizes the research findings and methodology.

Keywords Numerical Methods · PDEs · Discrete Geometry

1 Introduction

Welcome to the lightweight LaTeX collaboration environment. Start typing here. Press `Ctrl+Alt+B` (or `Cmd+Opt+B` on Mac) to manually compile your document.

2 Methodology

To solve the heat equation, we apply a finite difference method over a discrete mesh grid. We rely on the foundational theorems established by Smith and Doe [2023].

3 Results

The results of the simulation demonstrate the stability of the numerical method.

References

John Smith and Jane Doe. Numerical approaches to discrete differential geometry. *Journal of Applied Mathematics*, 42 (4):112–130, 2023.

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